On the triviality of inhomogeneous deformations of $\mathfrak{osp}(1|2n)$

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Abstract

We analyze the triviality of inhomogeneous γ-deformations of the oscillator Lie superalgebra $B(0,n) = \mathrm{osp}(1|2n)$ [3]. As the main theorem, we show that for $n \geq 2$, the γ-deformation is trivial if and only if all deformation parameters vanish. The proof is based on the explicit construction of $2n$ certificates (left null space vectors c satisfying $c^\top A_\mu = 0$ and $c^\top L_\mu \neq 0$) for the structure constant matrices A_μ of the coboundary operator. We provide a unified construction of certificates classified into three Families, and in particular clarify the geometric meaning of the coefficient $1 + \delta_{n,2}$ that appears in the Family III certificate. We also discuss the contrast with the exceptional case of $B(0,1) = \mathrm{osp}(1|2)$ (where all deformations are trivial). As an appendix, we outline the computational verification performed using exact rational arithmetic over $\mathbb{Q}$.

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1 Introduction

The deformation theory of Lie superalgebras is an important area of research in mathematical physics and representation theory. In [6, 10], the deformation theory of algebraic structures, including Lie algebras, was developed, and later the Chevalley–Eilenberg cohomology framework was extended to Lie superalgebras by Scheunert and Zhang [12]; see also [8, 5] for foundational work on cohomology of Lie superalgebras and infinite-dimensional Lie

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algebras. Among the basic Lie superalgebras classified by Kac [7], the orthosymplectic series $\mathrm{osp}(1|2n)$ plays a distinguished role: $\mathrm{osp}(1|2n)$ is the unique basic Lie superalgebra whose representation theory is completely reducible, mirroring the classical properties of semisimple Lie algebras. Moreover, $\mathrm{osp}(1|2n)$ arises naturally in mathematical physics, including oscillator realizations in supersymmetric quantum mechanics and the structure theory of W-algebras [4].

In [3], Bakalov and Sullivan study inhomogeneous deformations of Lie superalgebras through the framework of inhomogeneous bilinear forms. They develop oscillator Lie superalgebras obtained from inhomogeneous bilinear forms, focusing on the case of $\mathrm{osp}(1|2)$, and construct a concrete example in which the corresponding 2-cocycle γ is a coboundary, i.e., the deformation is trivial. Though their construction can be applied to general Lie superalgebras arising from oscillator realizations, the relation between the inhomogeneous bilinear forms and the triviality of deformations is not fully understood for other cases.

In this paper, we give a complete answer for the Lie superalgebras $B(0,n) = \mathrm{osp}(1|2n)$ $(n \geq 1)$. We show that for $n \geq 2$, the γ-deformation of $\mathrm{osp}(1|2n)$ is trivial if and only if all deformation parameters vanish. In contrast, we prove that all γ-deformations of $\mathrm{osp}(1|2)$ are trivial regardless of the values of the deformation parameters. This sharp dichotomy reflects a structural difference in the coboundary operator: for $n \geq 2$, the existence of certificates forces all deformation parameters to vanish, whereas for $n = 1$ the image of the coboundary map already contains every deformation direction.

Our proof of the main theorem is based on the "certificate method", which provides a systematic way to construct witnesses of non-triviality for each deformation parameter. We also adopt computational approaches with AI-assisted methods to verify the rank of the structure constant matrices and to discover patterns in the certificates, which lead to a unified construction independent of n for $n \geq 2$. Refer to §4 for details. This work is an example of collaboration between AI and human researchers, and the methodology is described in detail in [2].

The organization of the paper is as follows. In §2, we summarize the necessary definitions and notations related to the Lie superalgebra $\mathrm{osp}(1|2n)$, including its root system, oscillator realization, and central extension. In §3, we analyze the structure of the coboundary space and derive a dimension formula for the f-variable space in each weight sector. In §4, we construct explicit certificates for each deformation parameter. In §5, we state and prove the main theorem on triviality, and discuss the exceptional case of $B(0,1)$ and a conjecture for general $B(m,n)$. Finally, in Appendix A, we outline the computational verification performed using exact rational arithmetic.

2 Definitions and notations

We first summarize the basic facts about the Lie superalgebra $B(0,n) = \mathrm{osp}(1|2n)$ related to the root system and oscillator realization and central extension. For details, see [7, 4, 3]; for general background on Lie superalgebras we also refer to [9]. Our framework follows [3], but we adopt the oscillator notation of Frappat et al. [4], which differs in several conventions.

2.1 Root system of the Lie superalgebra $\mathrm{osp}(1|2n)$

It is well-known that $B(0,n) = \mathrm{osp}(1|2n)$ is one of the basic Lie superalgebras in the classification of Kac [7], consisting of the even subalgebra $\mathfrak{g}_{\bar{0}} \cong \mathrm{sp}(2n)$ and the odd part $\mathfrak{g}_{\bar{1}}$ (the fundamental representation of $\mathrm{sp}(2n)$). Let $\{e_1, \ldots, e_n\}$ be the standard orthonormal basis of $\mathfrak{h}^*$.

Definition 2.1 (Root system of $\mathrm{osp}(1|2n)$)**.** The root system of $\mathrm{osp}(1|2n)$ consists of the following:

$$\Delta_{\bar{0}}^+ = \{2\delta_i \mid 1 \le i \le n\} \cup \{\delta_i \pm \delta_j \mid 1 \le i < j \le n\} \qquad (|\Delta_{\bar{0}}^+| = n^2), \tag{1}$$

$$\Delta_{\bar{1}}^+ = \{\delta_i \mid 1 \le i \le n\} \qquad (|\Delta_{\bar{1}}^+| = n), \tag{2}$$

where $\delta_i = e_i$ in our setting. The simple root system is given by $\Pi = \{\alpha_k = e_k - e_{k+1} \mid k = 1, \ldots, n-1\} \cup \{\alpha_n = e_n\}$.

The basis of $\mathfrak{g}$ consists of the following:

- Cartan elements H_k ($k = 1, \ldots, n$): $H_k = h_{\alpha_k}$ (coroot of α_k),
- Even root vectors E_α ($\alpha \in \Delta_{\bar{0}}$),
- Odd root vectors $E_{\delta_j}, E_{-\delta_j}$ ($j = 1, \ldots, n$, roots $\pm\delta_j$).

The superdimension of $\mathfrak{g}$ is $\dim(\mathfrak{g}) = (2n^2 + n) \,|\, 2n$, with total dimension $2n^2 + 3n$.

Lemma 2.2 (Eigenvalues of Cartan elements)**.** *For $\alpha_k = e_k - e_{k+1}$ ($k < n$) or $\alpha_n = e_n$, the corresponding Cartan element H_k acts on a root vector E_α of root α with eigenvalue*

$$[H_k, E_\alpha] = \langle \alpha_k, \alpha \rangle \cdot E_\alpha,$$

where $\langle \cdot, \cdot \rangle$ is the bilinear form defined by $\langle e_i, e_j \rangle = \delta_{ij}$ (including coroot normalization). In particular,

$$[H_k, E_{\delta_j}] = (\delta_{jk} - \delta_{j,k+1}) \cdot E_{\delta_j} \quad (k < n), \qquad [H_n, E_{\delta_j}] = \delta_{jn} \cdot E_{\delta_j}. \tag{3}$$

2.2 Central extension and deformation parameters

The superalgebra $\mathrm{osp}(1|2n)$ is realized through the oscillator algebra. Using the auxiliary fermion a_0 (parity $p(a_0) = 1$, $a_0^2 = \frac{1}{2}$) and boson pairs $b_j^\pm$ ($j = 1, \ldots, n$, $[b_j^+, b_k^-] = \delta_{jk}$), $\mathrm{osp}(1|2n)$ is generated by the following elements:

- Cartan elements H_k $(k = 1, \ldots, n)$: $H_k = h_{\alpha_k}$,
- Even root vectors $E_{\delta_i - \delta_j} = b_i^+ b_j^-$ $(i < j)$, $E_{\delta_i + \delta_j} = b_i^+ b_j^+$ $(i < j)$, $E_{-2\delta_j} = b_j^- b_j^-$, etc.
- Odd root vectors $E_{\delta_j} = a_0 b_j^+$, $E_{-\delta_j} = a_0 b_j^-$ $(j = 1, \ldots, n)$.

Consider the central extension $\tilde{\mathfrak{g}} = \mathfrak{g} \oplus \mathbb{R}\kappa$ by an odd central element κ $(p(\kappa) = 1,\ \kappa^2 = 0)$ of the Lie superalgebra $\mathfrak{g}$. The extended bracket is defined using a 2-cocycle $\gamma\colon \mathfrak{g} \otimes \mathfrak{g} \to \mathfrak{g}$ as

$$[X, Y]_\gamma = [X, Y]_0 + \kappa \cdot \gamma(X, Y)$$

This is a general construction corresponding to an element of the Chevalley–Eilenberg cohomology $H^2(\mathfrak{g}, \mathfrak{g})$ (with coefficients in the adjoint module).

In the oscillator realization, γ is concretely parametrized through a modification of a_0^2:

$$a_0^2 \longmapsto \frac{1}{2} + \kappa \sum_{j=1}^{n} \sum_{s \in \{+,-\}} \mathrm{gb}_{a_0, b_j^s} \cdot b_j^s, \tag{4}$$

where gb_{a_0, b_j^s} are the **deformation parameters** ($2n$ in total). γ is a linear function of gb, and since κ is odd, γ is an odd 2-cocycle $(p(\gamma(X, Y)) = p(X) + p(Y) + 1)$.

Remark 2.3 (Parity and codomain conventions). We work over a base field F of characteristic 0 (concretely $F = \mathbb{Q}$) and use odd deformation symbols gb_{a_0, b_j^s} (parity 1). Then $\kappa \cdot \mathrm{gb}_{a_0, b_j^s} \cdot b_j^s$ has parity $1+1+0 = 0$, which is consistent with the parity of a_0^2. Accordingly, γ is an odd 2-cocycle $(p(\gamma(X, Y)) = p(X) + p(Y) + 1)$, and the extended bracket $[X, Y]_\gamma = [X, Y]_0 + \kappa\, \gamma(X, Y)$ preserves total parity. For the matrix equations used in the proofs, we work componentwise in each parameter direction and keep only the scalar coefficients in F.

2.3 Coboundary operator and triviality

Definition 2.4 (Coboundary)**.** For an odd linear map $f : \mathfrak{g} \to \mathfrak{g}$ $(p(f(X)) = p(X) + 1)$, the coboundary $\delta f : \mathfrak{g} \otimes \mathfrak{g} \to \mathfrak{g}$ is defined by

$$(\delta f)(X, Y) = (-1)^{p(X)}[X, f(Y)] - (-1)^{(p(X)+1)p(Y)}[Y, f(X)] - f([X, Y]). \tag{5}$$

The oddness of f is required so that $p(\kappa f(X)) = p(X)$ holds for the even automorphism $\varphi = \mathrm{Id} + \kappa f$ on $\tilde{\mathfrak{g}}$.

Definition 2.5 (Triviality)**.** A γ-deformation is said to be **trivial** if there exists an odd linear map $f\colon \mathfrak{g} \to \mathfrak{g}$ such that $\delta f = \gamma$. This corresponds to the removal of the deformation by the even automorphism $\varphi = \mathrm{Id} + \kappa f$.

By weight decomposition, the triviality equation $\delta f = \gamma$ decomposes into independent weight sectors μ:

$$A_\mu \cdot \mathbf{f}_\mu = L_\mu \cdot \mathbf{gb}_\mu, \tag{6}$$

where

- $\mathbf{f}_\mu \in F^{d_\mu}$: the f-variable vector of weight μ ($d_\mu = \dim C^1_\mu$, F is a field of characteristic 0),
- A_μ: the structure constant matrix (derived from brackets),
- L_μ: the γ-structure matrix (derived from deformation parameters),
- $\mathbf{gb}_\mu$: the gb parameters involved in weight μ.

Each weight sector $\mu = \pm e_j$ ($j = 1, \ldots, n$) contains the corresponding gb parameter $\mathrm{gb}_{a_0, b_j^\pm}$ and is related to the others by the action of Weyl group $W(B_n)$ (the hyperoctahedral group consisting of sign changes and permutations). By definition, each sector μ contains at most one gb parameter. So in our certificate argument for non-triviality (§4), each condition $c^\top L_\mu \neq 0$ suffices to force the corresponding gb parameter to vanish. In the general case where $\mathbf{gb}_\mu$ is a vector, the stronger condition $\mathrm{rank}([A_\mu \mid L_\mu]) > \mathrm{rank}(A_\mu)$ is required.

3 Structure of the coboundary space

3.1 Dimension formula

Proposition 3.1 (Dimension formula)**.** *For $n \geq 1$, the dimension of the f-variable space in any weight sector $\mu = \pm e_j$ of $B(0, n)$ is*

$$\dim C^1_\mu = 6n - 2.$$

Proof. Since the Weyl group $W(B_n)$ acts transitively on the set of short roots $\{\pm e_j\}_{j=1,\ldots,n}$, it suffices to take $\mu = e_1$ as a representative.

The f-variables $f(X \to Z) := f(X)\big|_Z$ (the Z-component of $f(X)$) correspond to pairs (X, Z) of basis elements satisfying $\mathrm{wt}(Z) - \mathrm{wt}(X) = \mu$. Based on root types, they are classified into the following 8 categories (for $\mu = e_1$):

Input X type	Output Z type	Correspondence	Count
H_k	E_{δ_1}	$e_1 - 0 = e_1$	n
$E_{-\delta_1}$	H_k	$0 - (-e_1) = e_1$	n
$E_{-\delta_j}$ $(j \neq 1)$	$E_{\delta_1 - \delta_j}$	$(e_1 - e_j) - (-e_j) = e_1$	$n - 1$
E_{δ_j} $(j \neq 1)$	$E_{\delta_1 + \delta_j}$	$(e_1 + e_j) - e_j = e_1$	$n - 1$
$E_{-2\delta_1}$	$E_{-\delta_1}$	$(-e_1) - (-2e_1) = e_1$	1
E_{δ_1}	$E_{2\delta_1}$	$2e_1 - e_1 = e_1$	1
$E_{\delta_j - \delta_1}$ $(j \neq 1)$	E_{δ_j}	$e_j - (e_j - e_1) = e_1$	$n - 1$
$E_{-\delta_1 - \delta_j}$ $(j \neq 1)$	$E_{-\delta_j}$	$(-e_j) - (-e_1 - e_j) = e_1$	$n - 1$

So we get the total $n+n+(n-1)+(n-1)+1+1+(n-1)+(n-1) = 6n-2$. (In rows 1–2, $\mathrm{wt}(H_k) = 0$ for all $k = 1, \ldots, n$, so all n Cartan elements contribute regardless of the value of k.) □

3.2 Rank and kernel

Remark 3.2 (Lower bound on the corank). Since the certificates c constructed in §4 satisfy $c^\top A_\mu = 0$, we have $\mathrm{corank}(A_\mu) \geq 1$ in each weight sector. That is, $\dim \mathrm{Ker}(\delta_\mu) \geq 1$, so the f-variables have at least one degree of freedom. The concrete value of the rank ($\mathrm{rank}(A_\mu) = 6n - 3$) has been computationally verified for $n = 2, 3, 4, 5$ in Appendix A. We emphasize that this exact rank formula is *not* used in the proof of the main theorem (Theorem 5.1); the certificate method requires only the existence of a non-trivial left null vector c with $c^\top L \neq 0$.

4 Construction of certificates

4.1 The certificate method

Definition 4.1 (Certificate)**.** A **certificate** (witness of non-triviality) for a weight sector μ is a vector c in the left null space of the matrix A_μ in equation (6) that satisfies $c^\top L_\mu \neq 0$.

Remark 4.2 (On the terminology). "Certificate" is the terminology adopted in this paper. It has essentially the same structure as an infeasibility certificate in the sense of Farkas' lemma in linear programming [11]. The conditions $c^\top A = 0$ and $c^\top L \neq 0$ provide, in the present per-sector one-parameter setting, a dual proof (obstruction) that the system $Ax = Lg$ has no solution for $g \neq 0$. As we mention above, each condition $c^\top L_\mu \neq 0$ only concerns a single parameter in gb. In general situations, we need to check the condition $\mathrm{rank}([A_\mu \mid L_\mu]) > \mathrm{rank}(A_\mu)$.

Lemma 4.3 (Non-triviality via certificates)**.** *If a certificate c for the weight sector μ exists and satisfies $c^\top L_\mu = \lambda \cdot \mathrm{gb}_i$ ($\lambda \neq 0$), then $\mathrm{gb}_i = 0$ is necessary for a trivial deformation (in the sense of Definition 2.5).*

Proof. When $\delta f = \gamma$, i.e., $A_\mu \mathbf{f}_\mu = L_\mu \mathbf{gb}_\mu$ holds, from $c^\top A_\mu = 0$ we get $0 = c^\top A_\mu \mathbf{f}_\mu = c^\top L_\mu \mathbf{gb}_\mu = \lambda \cdot \mathrm{gb}_i$. Since $\lambda \neq 0$, we conclude $\mathrm{gb}_i = 0$. □

In what follows, we construct a certificate for each of the $2n$ deformation parameters. Each certificate is a linear combination of 3 or 4 components of the form $c_k \cdot (\delta f)(X_k, Y_k)\big|_{Z_k}$. We will divide the certificates into three Families (I, II, III) based on their structure.

4.2 Family I certificates ($j = 1, \ldots, n-1$)

Proposition 4.4 (Family I)**.** *For $n \geq 2$ and $1 \leq j \leq n-1$, the certificate c_j^+ for gb_{a_0,b_j^+} is defined by the following 4 components.*

***Case** $j = 1$:*

Component	$(X_k,\ Y_k)$	Z_k	*Coefficient* c_k
1	$(H_1,\ E_{2\delta_1})$	E_{δ_1}	$+1$
2	$(H_1,\ E_{\delta_1})$	H_2	$+2$
3	$(H_1,\ E_{-\delta_1})$	$E_{-2\delta_1}$	-4
4	$(E_{2\delta_1},\ E_{-\delta_1})$	H_2	$+1$

***Case** $j \geq 2$:*

Component	$(X_k,\ Y_k)$	Z_k	*Coefficient* c_k
1	$(H_{j-1},\ E_{2\delta_j})$	E_{δ_j}	-1
2	$(H_{j-1},\ E_{\delta_j})$	H_{j+1}	-2
3	$(H_{j-1},\ E_{-\delta_j})$	$E_{-2\delta_j}$	$+4$
4	$(E_{2\delta_j},\ E_{-\delta_j})$	H_{j+1}	$+1$

These satisfy:

$$c_j^{+\top} A_\mu = 0, \qquad c_j^{+\top} L_\mu = 4 \cdot \mathrm{gb}_{a_0,b_j^+}.$$

4.3 Family II certificates ($j = 1, \ldots, n-1$)

Proposition 4.5 (Family II)**.** *For $n \geq 2$ and $1 \leq j \leq n-1$, define the certificate c_j^- for gb_{a_0,b_j^-} as follows:*

For $j = 1$ (3 components):

$$\begin{aligned} c_1^- &= (-2) \cdot (\delta f)(H_1, E_{-\delta_1})\big|_{H_1} + (+2) \cdot (\delta f)(H_1, E_{-\delta_1})\big|_{H_2} \\ &\quad + (+1) \cdot (\delta f)(E_{2\delta_2}, E_{-\delta_1})\big|_{E_{2\delta_2}}. \end{aligned} \tag{7}$$

For $j \geq 2$ (3 components):

$$\begin{aligned} c_j^- &= (+2) \cdot (\delta f)(H_{j-1}, E_{-\delta_j})\big|_{H_j} + (-2) \cdot (\delta f)(H_{j-1}, E_{-\delta_j})\big|_{H_{j+1}} \\ &\quad + (+1) \cdot (\delta f)(E_{2\delta_{j+1}}, E_{-\delta_j})\big|_{E_{2\delta_{j+1}}}. \end{aligned} \tag{8}$$

These satisfy:

$$c_j^{-\top} A_\mu = 0, \qquad c_j^{-\top} L_\mu = 2 \cdot \mathrm{gb}_{a_0,b_j^-}.$$

To prove these propositions, we will show that these properties are independent of n.

4.4 n-invariance

Lemma 4.6 (n-invariance of Family I/II)**.** *The Family I/II certificates are independent of n for fixed j in the range $n \geq j+1$. That is, the set of f-variables constituting $c^\top A = 0$, their coefficients, and the value of $c^\top L$ are all invariant with respect to n.*

Proof. For Families I and II, the generators appearing in each certificate belong to the j-th slot set S_j (defined below). The brackets $[X, Y]$ ($X, Y \in S_j$) within S_j depend only on the eigenvalues $\langle \alpha_k, \alpha \rangle$ for $k = j-1, j, j+1$ and the e_j, e_{j+1} components of α, and do not depend on n. In the expansion of (δf), the f-variables $f(W \to V)$ lie within the range $W, V \in S_j$, so under the condition $n \geq j+1$, all structure constants are determined and independent of n. □

By this lemma, to verify the certificate identities of Propositions 4.4 and 4.5, it suffices to check a single value of n satisfying $n \geq j+1$ for each j. In the proofs below, we verify the cases $j = 1$ (using $n = 2$) and $j \geq 2$ (using $n = j+1$), from which the identities hold for all $n \geq j+1$.

Proof of Proposition 4.4. **(i) Proof that $c^\top A = 0$.**

The f-variables appearing in the expansion of $c^\top A$ are restricted to the j-th slot set

$$S_j = \begin{cases} \{H_1,\ H_2,\ E_{2\delta_1},\ E_{-2\delta_1},\ E_{\delta_1},\ E_{-\delta_1}\} & (j = 1), \\ \{H_{j-1},\ H_{j+1},\ E_{2\delta_j},\ E_{-2\delta_j},\ E_{\delta_j},\ E_{-\delta_j}\} & (j \geq 2), \end{cases}$$

and by n-invariance (Lemma 4.6), it suffices to verify the two cases below.

Case $j = 1$: The four f-variables contribute as follows:

f-variable	Comp. 1	Comp. 2	Comp. 3	Comp. 4	Total
$f(E_{2\delta_1} \to E_{\delta_1})$	-1			$+1$	0
$f(H_1 \to E_{-\delta_1})$	$+2$	$+2$	-4		0
$f(E_{\delta_1} \to H_2)$		-2		$+2$	0
$f(E_{-\delta_1} \to E_{-2\delta_1})$			$+4$	-4	0

Case $j \geq 2$: The sign of the Cartan eigenvalue flips: $[H_1, E_{\delta_1}] = +E_{\delta_1}$ but $[H_{j-1}, E_{\delta_j}] = -E_{\delta_j}$ for $j \geq 2$ (see Lemma 2.2). This reverses the sign of the $f(H_{j-1} \to E_{-\delta_j})$ row:

f-variable	Comp. 1	Comp. 2	Comp. 3	Comp. 4	Total
$f(E_{2\delta_j} \to E_{\delta_j})$	-1			$+1$	0
$f(H_{j-1} \to E_{-\delta_j})$	-2	-2	$+4$		0
$f(E_{\delta_j} \to H_{j+1})$		-2		$+2$	0
$f(E_{-\delta_j} \to E_{-2\delta_j})$			$+4$	-4	0

(ii) Proof that $c^\top L = 4 \cdot \mathrm{gb}_{a_0,b_j^+}$.

Contributions to $c^\top L$ arise only from components whose brackets involve even–odd pairs with a γ-deformation term.

- Component 1: $(H_{j(\pm 1)}, E_{2\delta_j})$ is an even–even pair $\to$ no γ $\to$ contribution 0.
- Component 2: $(H_{j(\pm 1)}, E_{\delta_j})$ is an even–odd pair $\to$ γ present. Since $\gamma(H_{j(\pm 1)}, E_{\delta_j})\big|_{H_{j\pm 1}} = (\pm 2) \cdot \mathrm{gb}_{a_0,b_j^+}$, this component contributes $c_2 \cdot (\pm 2) \cdot \mathrm{gb}_{a_0,b_j^+} = (\pm 2)(\pm 2) \cdot \mathrm{gb}_{a_0,b_j^+} = 4 \cdot \mathrm{gb}_{a_0,b_j^+}$.
- Component 3: $(H_{j(\pm 1)}, E_{-\delta_j})$ is an even–odd pair $\to$ γ present, but $\gamma(H_{j(\pm 1)}, E_{-\delta_j})\big|_{E_{-2\delta_j}}$ contains gb_{a_0,b_j^-}, not gb_{a_0,b_j^+} $\to$ contribution 0.
- Component 4: $(E_{2\delta_j}, E_{-\delta_j})$ is an even–odd pair, but $\gamma(E_{2\delta_j}, E_{-\delta_j})\big|_{H_{j\pm 1}}$ has no gb_{a_0,b_j^+} component in the $H_{j\pm 1}$ direction $\to$ contribution 0.

Hence

$$c^\top L = 4 \cdot \mathrm{gb}_{a_0,b_j^+}.$$

By Lemma 4.6, the above identities hold for all $n \geq j+1$; they have been confirmed by direct calculation[1] for $n = 2$. □

Proof of Proposition 4.5. **(i) Proof that $c^\top A = 0$.**

As with Family I, the f-variables are restricted to the j-th slot set S_j and possess n-invariance (Lemma 4.6).

Case $j = 1$: The three f-variables contribute as follows:

f-variable	Comp. 1	Comp. 2	Comp. 3	Total
$f(E_{-\delta_1} \to H_1)$	-2		$+2$	0
$f(H_1 \to E_{\delta_1})$	-2	$+2$		0
$f(E_{-\delta_1} \to H_2)$		$+2$	-2	0

Case $j \geq 2$: By the Cartan eigenvalue sign flip $[H_1, E_{-\delta_1}] = -E_{-\delta_1}$ vs. $[H_{j-1}, E_{-\delta_j}] = +E_{-\delta_j}$ for $j \geq 2$, the $f(H_{j-1} \to E_{\delta_j})$ row reverses sign:

f-variable	Comp. 1	Comp. 2	Comp. 3	Total
$f(E_{-\delta_j} \to H_j)$	-2		$+2$	0
$f(H_{j-1} \to E_{\delta_j})$	$+2$	-2		0
$f(E_{-\delta_j} \to H_{j+1})$		$+2$	-2	0

(ii) Proof that $c^\top L = 2 \cdot \mathrm{gb}_{a_0,b_j^-}$.

Contributions to $c^\top L$ arise only from components whose brackets involve even–odd pairs with a γ-deformation term.

[1] Computational verification using exact rational arithmetic; see Appendix.

- Component 1: $(H_{j(\pm 1)}, E_{-\delta_j})$ is an even–odd pair $\to \gamma$ present. Since $\gamma(H_{j(\pm 1)}, E_{-\delta_j})\big|_{H_j} = (\pm 1) \cdot \mathrm{gb}_{a_0,b_j^-}$, this component contributes $(\pm 2) \cdot (\pm 1) \cdot \mathrm{gb}_{a_0,b_j^-} = 2 \cdot \mathrm{gb}_{a_0,b_j^-}$.
- Component 2: $(H_{j(\pm 1)}, E_{-\delta_j})$ is the same even–odd pair as component 1, but projected onto $H_{j\pm 1}$. The γ-output $\gamma(H_{j(\pm 1)}, E_{-\delta_j})$ has no gb_{a_0,b_j^-} component in the $H_{j\pm 1}$ direction $\to$ contribution 0.
- Component 3: $(E_{2\delta_{j+1}}, E_{-\delta_j})$ is an even–odd pair. The component

$$\gamma(E_{2\delta_{j+1}}, E_{-\delta_j})\big|_{E_{2\delta_{j+1}}}$$

  has no gb_{a_0,b_j^-} contribution, so this term contributes 0.

Hence

$$c^\top L = 2 \cdot \mathrm{gb}_{a_0,b_j^-}.$$

As in the proof of Proposition 4.4, by Lemma 4.6 the above identities hold for all $n \geq j+1$, and have been confirmed by direct calculation for $n = 2$. □

Note that the remaining case (Family III) does *not* enjoy n-invariance.

4.5 Family III certificates

Family III corresponds to $\mathrm{gb}_{a_0,b_n^\pm}$ (the last index), and since the generators used depend on n, it does not possess the n-invariance of Families I/II.

Lemma 4.7 (Cartan eigenvalue for Family III)**.** *The eigenvalue of H_{n-1} ($n \geq 2$) on E_{δ_1} and $E_{-\delta_1}$ is*

$$[H_{n-1}, E_{\delta_1}] = \delta_{n,2} \cdot E_{\delta_1}, \qquad [H_{n-1}, E_{-\delta_1}] = -\delta_{n,2} \cdot E_{-\delta_1}.$$

That is, the eigenvalue is ± 1 when $n = 2$ and 0 when $n \geq 3$.

Proof. $H_{n-1} = h_{\alpha_{n-1}}$ corresponds to the simple root $\alpha_{n-1} = e_{n-1} - e_n$. Since E_{δ_1} has root e_1 and $E_{-\delta_1}$ has root $-e_1$,

$$[H_{n-1}, E_{\delta_1}] = \langle e_{n-1} - e_n, e_1 \rangle \cdot E_{\delta_1} = \delta_{1,n-1} \cdot E_{\delta_1} = \delta_{n,2} \cdot E_{\delta_1},$$

and similarly $[H_{n-1}, E_{-\delta_1}] = \langle e_{n-1} - e_n, -e_1 \rangle \cdot E_{-\delta_1} = -\delta_{n,2} \cdot E_{-\delta_1}$. □

Proposition 4.8 (Family III certificate)**.** *For $n \geq 2$, define the certificate c_n^- for gb_{a_0,b_n^-} by the following 4 components:*

Component	(X_k, Y_k)	*Output* Z_k	*Coeff.* c_k
1	$(H_{n-1},\ E_{\delta_1-\delta_n})$	$E_{-\delta_1}$	1
2	$(H_{n-1},\ E_{\delta_1})$	$E_{\delta_1+\delta_n}$	$1+\delta_{n,2}$
3	$(H_{n-1},\ E_{-\delta_n})$	H_2	-1
4	$(E_{\delta_1-\delta_n},\ E_{\delta_1})$	H_2	1

Similarly, define the certificate c_n^+ for gb_{a_0,b_n^+} by the following 4 components:

Component	(X_k, Y_k)	*Output* Z_k	*Coeff.* c_k
1	$(H_{n-1},\ E_{\delta_1+\delta_n})$	E_{δ_1}	-1
2	$(H_{n-1},\ E_{-\delta_1})$	$E_{\delta_1-\delta_n}$	$1+\delta_{n,2}$
3	$(H_{n-1},\ E_{\delta_n})$	H_2	-1
4	$(E_{\delta_1+\delta_n},\ E_{-\delta_1})$	H_2	1

These satisfy:

$$c_n^{\pm\top} A_\mu = 0, \qquad c_n^{\pm\top} L_\mu = 1 \cdot \mathrm{gb}_{a_0,b_n^\pm}.$$

Proof. We give the proof for c_n^-; the proof for c_n^+ is entirely analogous, with the exchanges $E_{\delta_1} \leftrightarrow E_{-\delta_1}$, $E_{\delta_n} \leftrightarrow E_{-\delta_n}$, and $E_{\delta_1+\delta_n} \leftrightarrow E_{\delta_1-\delta_n}$. By Lemma 4.7, the component involving $E_{-\delta_1}$ in c_n^+ receives the same correction factor $1+\delta_{n,2}$, and the cancellation argument proceeds identically.

(i) Proof that $c^\top A = 0$.
We consider cases according to the value of n.

Case 1: $n \geq 3$. By Lemma 4.7, $[H_{n-1}, E_{\delta_1}] = 0$, so no extra f-variables arise from the expansion of component 2. The f-variables appearing in $c^\top A$ are the following 4 only:

f-variable	Comp. 1	Comp. 2	Comp. 3	Comp. 4	Total
$f(E_{\delta_1-\delta_n} \to E_{-\delta_1})$	-1			$+1$	0
$f(E_{\delta_1} \to E_{\delta_1+\delta_n})$		-1		$+1$	0
$f(E_{-\delta_n} \to H_2)$			$+1$	-1	0
$f(H_{n-1} \to E_{\delta_n})$	-1	$+1$			0

Each f-variable contributes to exactly 2 components with opposite signs, so they cancel completely.

Case 2: $n = 2$. $[H_1, E_{\delta_1}] = +E_{\delta_1} \neq 0$, but the correction $c_2 = 1+\delta_{2,2} = 2$ ensures that the f-variables still number 4 and all cancel:

f-variable	Comp. 1	Comp. 2	Comp. 3	Comp. 4	Total
$f(E_{\delta_1-\delta_2} \to E_{-\delta_1})$	-1			$+1$	0
$f(E_{\delta_1} \to E_{\delta_1+\delta_2})$		-2		$+2$	0
$f(E_{-\delta_2} \to H_2)$			$+1$	-1	0
$f(H_1 \to E_{\delta_2})$	-1	$+2$	-1		0

In the fourth row $f(H_1 \to E_{\delta_2})$, the contribution $+2$ from component 2 with coefficient $c_2 = 2$ exactly cancels the contributions $-1, -1$ from components 1 and 3.

(ii) Proof that $c^\top L = \mathrm{gb}_{a_0,b_n^-}$.

Contributions to $c^\top L$ arise only from components whose brackets involve even–odd pairs with a γ-deformation term.

- Component 1: $(H_{n-1}, E_{\delta_1-\delta_n})$ is an even–even pair $\to$ no γ $\to$ contribution 0.
- Component 2: (H_{n-1}, E_{δ_1}) is an even–odd pair $\to$ γ present.
$$(1+\delta_{n,2}) \cdot \gamma(H_{n-1}, E_{\delta_1})\big|_{E_{\delta_1+\delta_n}} = (1+\delta_{n,2}) \cdot \mathrm{gb}_{a_0,b_n^-}.$$
- Component 3: $(H_{n-1}, E_{-\delta_n})$ is an even–odd pair $\to$ γ present. The component
$$\gamma(H_{n-1}, E_{-\delta_n})\big|_{H_2}$$
contains gb_{a_0,b_n^-} only when $H_2 = H_n$, i.e., only when $n = 2$:
$$(-1) \cdot \gamma(H_{n-1}, E_{-\delta_n})\big|_{H_2} = -\delta_{n,2} \cdot \mathrm{gb}_{a_0,b_n^-}.$$
- Component 4: $(E_{\delta_1-\delta_n}, E_{\delta_1})$ is an even–odd pair, but the γ output has no gb_{a_0,b_n^-} component in the H_2 direction $\to$ contribution 0.

Summing up:
$$c^\top L = (1+\delta_{n,2}) \cdot \mathrm{gb}_{a_0,b_n^-} + (-\delta_{n,2}) \cdot \mathrm{gb}_{a_0,b_n^-} = 1 \cdot \mathrm{gb}_{a_0,b_n^-}.$$
This holds for all $n \geq 2$. □

Remark 4.9 (Geometric meaning of the unified coefficient). In $c_2 = 1 + \delta_{n,2}$, we have $\delta_{n,2} = \langle e_{n-1} - e_n, e_1 \rangle$, which is naturally determined by root geometry. When $n = 2$, $e_{n-1} = e_1$ so H_{n-1} acts non-trivially on E_{δ_1}, whereas for $n \geq 3$ it acts trivially by orthogonality. This coefficient precisely compensates for the additional γ-contribution (the $-\delta_{n,2}$ from component 3) that is specific to $n = 2$, thereby preserving $c^\top L = 1$.

5 Main theorem and remarks

Theorem 5.1 (Triviality criterion for B(0,n))**.** *For $n \geq 2$, the γ-deformation of $B(0,n) = \mathrm{osp}(1|2n)$ is trivial if and only if all deformation parameters vanish (in the sense of Definition 2.5):*
$$\mathrm{gb}_{a_0,b_j^+} = \mathrm{gb}_{a_0,b_j^-} = 0 \quad (j = 1, \ldots, n).$$

Proof. ($\Leftarrow$): If gb $= 0$, then $\gamma = 0 = \delta(0)$, so the deformation is trivial.

($\Rightarrow$): Assume the γ-deformation is trivial, i.e., there exists f satisfying $\delta f = \gamma$. By Propositions 4.4, 4.5, and 4.8, certificates exist for each gb parameter:

Parameter	Certificate	$c^\top L$
gb_{a_0,b_j^+} $(j=1,\dots,n-1)$	Family I: c_j^+	$4\cdot\mathrm{gb}_{a_0,b_j^+}$
gb_{a_0,b_j^-} $(j=1,\dots,n-1)$	Family II: c_j^-	$2\cdot\mathrm{gb}_{a_0,b_j^-}$
gb_{a_0,b_n^+}	Family III$^+$: c_n^+	$1\cdot\mathrm{gb}_{a_0,b_n^+}$
gb_{a_0,b_n^-}	Family III$^-$: c_n^-	$1\cdot\mathrm{gb}_{a_0,b_n^-}$

Applying Lemma 4.3 to each certificate, we obtain $\mathrm{gb}_{a_0,b_j^\pm}=0$ for all $2n$ parameters. □

In contrast, for $n=1$, we can directly prove that the deformation is always trivial:

Proposition 5.2 (Complete triviality of $B(0,1)$)**.** *For $B(0,1)=\mathrm{osp}(1|2)$, the γ-deformation is trivial for any value of the* gb *parameters.*

Proof. We verify the condition $\mathrm{Im}(L)\subseteq\mathrm{Im}(A)$, which is equivalent to the existence of a solution f to $\delta f=\gamma(\mathrm{gb})$ for all gb. This condition is characterized by

$$\mathrm{rank}([A\mid L])=\mathrm{rank}(A).$$

By exact computation over $\mathbb{Q}$ using the structure constants of $B(0,1)$ (see Appendix A), we obtain:

$$\dim C^1=12,\quad \mathrm{rank}(A)=10,\quad \mathrm{rank}([A\mid L])=10.$$

Hence $\mathrm{Im}(L)\subseteq\mathrm{Im}(A)$, and for each basis direction of gb, a particular solution f can be constructed explicitly:

$$\begin{aligned}
&\text{for } \mathrm{gb}_{a_0,b_1^+}\neq 0:\quad f(H_1)=-\mathrm{gb}_{a_0,b_1^+}\cdot E_{-\delta_1},\quad f(E_{2\delta_1})=-2\,\mathrm{gb}_{a_0,b_1^+}\cdot E_{\delta_1},\\
&\text{for } \mathrm{gb}_{a_0,b_1^-}\neq 0:\quad f(H_1)=-\mathrm{gb}_{a_0,b_1^-}\cdot E_{\delta_1},\quad f(E_{-2\delta_1})=-2\,\mathrm{gb}_{a_0,b_1^-}\cdot E_{-\delta_1},
\end{aligned}$$

with all other components of f equal to zero. One verifies directly that $\delta f=\gamma(\mathrm{gb})$ for general $\mathrm{gb}=\mathrm{gb}_{a_0,b_1^+}\,e^+ + \mathrm{gb}_{a_0,b_1^-}\,e^-$, where $e^+=(1,0)$ and $e^-=(0,1)$ are the canonical basis vectors of the two-dimensional parameter space $\{(\mathrm{gb}_{a_0,b_1^+},\mathrm{gb}_{a_0,b_1^-})\}$.

Note that $\dim\mathrm{Ker}(\delta_\mu)=2$ does not by itself imply the existence of a solution: it only measures non-uniqueness. The correct criterion used here is $\mathrm{rank}([A\mid L])=\mathrm{rank}(A)$. □

Corollary 5.3. *We have the following:*

$$\textit{The } \gamma\textit{-deformation of } B(0,n) \textit{ is trivial} \iff \begin{cases} \textit{always holds} & (n=1),\\ \mathrm{gb}=0 & (n\geq 2).\end{cases} \tag{9}$$

This dichotomy arises because for $n \geq 2$ the certificates force $\mathrm{gb} = 0$, whereas for $n = 1$ the condition $\mathrm{rank}([A \mid L]) = \mathrm{rank}(A)$ holds, meaning every γ lies in the image of δ.

Conjecture 5.4 (General $B(m,n)$)**.** *For $B(m,n) = \mathrm{osp}(2m+1|2n)$ with $m+n \geq 2$, the γ-deformation is trivial if and only if* $\mathrm{gb} = 0$.

This conjecture has been computationally verified for small m, n (by confirming $\mathrm{rank}([A \mid L]) = \mathrm{rank}(A) + \mathrm{rank}(L)$). For $m \geq 1$, the deformation parameters involve coefficients in $\mathbb{Q}(\sqrt{2})$ (due to the contributions from fermionic oscillators $a_j^{\pm}$), so generalizing the proof requires rank invariance over field extensions. This is guaranteed by the standard fact that rank is preserved under scalar extension: for a matrix M over a field F, $\mathrm{rank}_{F'}(M) = \mathrm{rank}_F(M)$ for any extension F'/F.

Human-AI collaboration

Our research was conducted through a structured human-AI collaborative workflow, leveraging large language models (e.g., GitHub Copilot) for computational verification, code generation, and drafting, while the design of proof strategies and mathematical validation were led by human researchers. Details of the methodology, including workflow design, role allocation, data pipeline, and lessons learned, are documented in a separate paper [2].

A Implementation of computational verification

The theorems and propositions in this paper were verified using computational methods implemented in Python. Below we outline the approach. We note that the computational artifacts for this work are available at [1], which includes a Jupyter notebook with the verification code and data files containing the structure constants and γ-structures for small values of n.

Coefficient field and exact arithmetic

To ensure exactness of the computations, the deformation parameters were treated as $\mathrm{gb}_{a_0, b_j^s} \in \mathbb{Q}$, and exact rational arithmetic was performed using Python's `fractions.Fraction`. Since the theoretical part of this paper (§3–§5) involves structure constants and certificate coefficients that are all integers, the results hold over any field F of characteristic 0.

Rank formula (computational verification)

The exact rank of the structure constant matrix A_μ (over $\mathbb{Q}$) was computed for $n = 1, \ldots, 5$, yielding the following:

n	$\dim C^1$	$\mathrm{rank}(A_\mu)$	$\dim \mathrm{Ker}(\delta_\mu)$
1	12*	10*	2*
2	10	9	1
3	16	15	1
4	22	21	1
5	28	27	1

*For $n = 1$, values are for the full coboundary matrix A (all weight sectors combined). For $n \geq 2$, values are per weight sector $\mu = \pm e_j$.

From this data, the following conjecture for $n \geq 2$ is suggested:

$$\mathrm{rank}(A_\mu) = 6n - 3, \qquad \dim \mathrm{Ker}(\delta_\mu) = 1.$$

The existence of certificates (§4) theoretically establishes $\mathrm{corank}(A_\mu) \geq 1$, but $\mathrm{corank}(A_\mu) \leq 1$ remains a computational observation. Note that the proof of the main theorem (Theorem 5.1) does not require this equality.

Verification environment

- **Hardware**: Apple M3 Ultra (28-core CPU, 512GB RAM)
- **Language**: Python 3.12
- **Numerical computation**: NumPy (numerical matrix rank computation)
- **Symbolic computation**: SymPy (exact rank computation, symbolic verification)
- **Rational arithmetic**: `fractions.Fraction` (Python standard library; exact computation of certificate coefficients and $c^\top A$, $c^\top L$)
- **Data format**: JSON (serialization of structure constants and γ-structures)

Verification items and corresponding scripts

Verification item	Corresponding proposition
Rank verification	Rank formula above
Certificate algebraic verification	Propositions 4.4–4.8
n-invariance verification	Lemma 4.6
Dimension formula verification	Proposition 3.1

Data pipeline

The structure constants are automatically generated from the computational rules of the oscillator algebra:

1. Realize the generators of $\mathrm{osp}(1|2n)$ in terms of oscillators and compute all brackets based on PBW (Poincaré–Birkhoff–Witt) canonical ordering.

2. Save the results in JSON format (`data/algebra_structures/`).
3. Save the γ-deformation structure (correction terms based on equation (4)) similarly in JSON (`data/gamma_structures/`).
4. Each verification script reads the JSON files and performs verification using exact rational arithmetic over $\mathbb{Q}$.

All $\sum_{n=2}^{5} 2n = 4+6+8+10 = 28$ certificates for $n = 2, 3, 4, 5$ (osp(1|4) through osp(1|10)) have been confirmed to satisfy $c^\top A = 0$ and $c^\top L \neq 0$.

Preliminary verification for general case (supplement for conjecture)

As evidence for Conjecture 5.4, we computationally verified the rank condition $\mathrm{rank}([A \mid L]) = \mathrm{rank}(A) + \mathrm{rank}(L)$ for types with $m \geq 1$, $n \geq 1$, and $m + n \leq 6$.

Type	osp	dim	size of gb	Computation time	Verdict
$B(1,1)$	osp(3|2)	12	6	$<$ 1s	PASS
$B(1,2)$	osp(3|4)	25	12	1.4s	PASS
$B(1,3)$	osp(3|6)	42	18	37.9s	PASS
$B(2,1)$	osp(5|2)	23	10	0.8s	PASS
$B(2,2)$	osp(5|4)	40	20	25.8s	PASS
$B(3,1)$	osp(7|2)	38	14	18.2s	PASS
$B(3,2)$	osp(7|4)	59	28	461s	PASS
$B(3,3)$	osp(7|6)	84	42	$\sim$1–3h (est.)	—

The $B(3,3)$ computation time is extrapolated from scaling analysis of the measured data for $B(m,n)$ with $m + n \leq 5$. The coboundary matrix A for $B(3,3)$ has shape 296352×3528, and the full SVD requires approximately 700 GB of memory, which exceeds available hardware (512 GB). A memory-efficient SVD implementation or distributed computation would be needed to complete this verification.

The verification script `verify_Bmn_rank.py` computes the rank of $[A \mid L]$ via SVD with a threshold of 0.1 on the smallest singular value. The computation is performed over $\mathbb{R}$ via embedding from $\mathbb{Q}(\sqrt{2})$, which preserves rank. An optional `--sympy` flag enables exact rank computation over algebraic number fields for cross-validation.

Reproducibility metadata. The verification artifacts (scripts and data) are available at [1], release tag `v1.0`. The exact command is `pythonscripts/verify_Bmn_rank.py--max-m3--max-n3`. For $B(1,1)$ and $B(2,1)$, exact-rank cross-checks via SymPy confirmed agreement with the SVD-based verdicts. For larger cases, the smallest non-zero singular value exceeds 1.0 (well above the threshold 0.1), providing robust rank decisions.

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